

PAVAN SAI CHILUKALA

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PROFESSIONAL SUMMARY

Full-stack and mobile software engineer with production experience building and shipping web, mobile, and AI-integrated applications end to end. Proficient across React, React Native, Swift, TypeScript, Python, Node.js, PostgreSQL, MongoDB, AWS, and CI/CD pipelines. Currently volunteering as iOS Developer at Streetcare, a non-profit helping homeless communities. Completed M.S. Computer Science (GPA 3.8, December 2025). Hands-on experience integrating OpenAI and Anthropic LLM APIs into production systems. Actively uses Claude Code, Cursor, and GitHub Copilot as core workflow tools. On OPT, open to relocate anywhere in the US. Available immediately.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Swift, Python, Java, C/C++, PHP, SQL
Mobile: React Native, SwiftUI, Expo, iOS (App Store), Android (Play Store), Core Audio (CATap)
Frontend: React, Next.js, Vite, Redux, Tailwind CSS, Bootstrap, HTML5/CSS3
Backend: Node.js, Express.js, Flask, FastAPI, REST API Design, RESTful API Integration
Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Neo4j, Firebase, Firestore
AI / ML: OpenAI API, Anthropic Claude API, LLM Integration, Prompt Engineering, PyTorch, NLP, LangGraph
AI Dev Tools: Claude Code, GitHub Copilot, Cursor, ChatGPT, Gemini
Visualization: D3.js, Plotly, Matplotlib, Tableau
Cloud & DevOps: AWS S3, GCP, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Vercel, Render
Tools: Git, GitHub, Postman, Xcode, VSCode, PyCharm, WebStorm, Linux (Ubuntu)

WORK EXPERIENCE

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| Streetcare
<i>iOS Developer Volunteer</i> | Remote
<i>Feb 2026 – Present</i> |
| <ul style="list-style-type: none">– Contributing as volunteer iOS developer for a non-profit helping homeless communities, adding new features and resolving bugs in a production Swift and SwiftUI application used by field workers.– Integrating Firestore as real-time cloud database backend, managing data synchronization between the iOS app and backend services for reliable, up-to-date field operations.– Performing technical root cause analysis, peer code reviews, and participating in the full SDLC: design, development, testing, and release. | |
| University of Central Missouri
<i>Graduate Teaching Assistant – Algorithms & Data Structures</i> | Warrensburg, MO
<i>Jan 2025 – Dec 2025</i> |
| <ul style="list-style-type: none">– Supported Design and Analysis of Algorithms course for 50+ undergraduate students; evaluated 500+ assignments for logical correctness, time complexity (Big O), and algorithmic efficiency.– Provided structured written feedback on pseudocode, flowcharts, and implementation quality; mentored students on optimal data structure selection and scalable system design.– Collaborated with faculty to resolve curriculum conflicts, develop course content, and deliver improvements with high ownership and minimal oversight. | |
| Dhanush Infotech Pvt Ltd
<i>React Native Developer Intern</i> | Hyderabad, India
<i>Sep 2022 – Dec 2022</i> |
| <ul style="list-style-type: none">– Built and shipped production iOS and Android mobile features across the full React Native stack, publishing 7 production app updates to both the App Store and Play Store.– Architected 50+ modular UI components; applied memoization achieving 80% faster turnaround across 20+ existing components; improved UI responsiveness by 70% via lazy async dynamic loading.– Integrated 100+ RESTful APIs with Redux state management; implemented Google OAuth, Firebase push notifications, and 20+ deep-linked navigation flows.– Designed and implemented CI/CD workflow via GitHub Actions; managed version control and resolved complex merge conflicts in a strict Git-flow environment. | |

PROJECTS

volumeMixer <i>Swift, SwiftUI, Core Audio (CATap), macOS</i> github/volumeMixer	2026
– Built a macOS menu bar utility for independent per-application volume control using low-level Core Audio Tap (CATap) APIs, working directly at the hardware abstraction layer with no third-party dependencies. – Designed from first principles and shipped as open-source, demonstrating deep Swift and macOS systems programming.	
AI Full-Stack Platform <i>React, TypeScript, Node.js, Express.js, PostgreSQL, AWS, OpenAI, Anthropic</i>	2025
– Built an end-to-end AI-integrated web application with OpenAI and Anthropic LLM APIs: streaming responses, prompt engineering, tool calls, error handling, and latency management in production. – Designed scalable REST API layer on Express.js with PostgreSQL backend; deployed on AWS with CI/CD pipelines maintaining 99.9% uptime. Used Claude Code and GitHub Copilot throughout.	
The Daily Wag <i>React, TypeScript, Next.js, Node.js, MySQL, AWS S3, Docker, JWT</i> github/daily-wag	2025
– Led full-stack architecture of a pet clinic consumer platform: modular React/TypeScript frontend, Node.js and PHP backend, MySQL database, JWT authentication, and AWS S3 cloud storage. – Engineered custom scheduling APIs preventing concurrent booking conflicts; built admin dashboard with purchase analytics across 200+ products and dynamic loyalty-based pricing logic. – Containerized with Docker and deployed via CI/CD pipelines with zero-downtime releases.	
C3-Carpool Connect <i>React, TypeScript, Python Flask, PostgreSQL, MongoDB</i> github/carpool-frontend	2024
– Developed a real-time ride-sharing platform matching users via optimized geo-spatial proximity queries, achieving sub-second matching latency across distributed PostgreSQL and MongoDB backends. – Maintained automated CI/CD pipelines on Vercel and Render ensuring 99.9% system uptime and seamless production feature rollouts.	
Black Friday Sales Prediction <i>Python, PyTorch, Machine Learning, NLP, D3.js, Plotly, Matplotlib, Tableau</i>	2022
– Developed a Python predictive ML model using PyTorch and NLP to analyze customer behavior patterns and optimize holiday inventory decisions from historical sales data. – Built interactive data visualization dashboards using D3.js, Plotly, Matplotlib, and Tableau to surface insights for non-technical stakeholders.	

EDUCATION

University of Central Missouri	Warrensburg, MO
<i>Master of Science, Computer Science GPA: 3.8 / 4.0</i>	<i>Jan 2024 – Dec 2025</i>
Jawaharlal Nehru Technological University (JNTU)	India
<i>Bachelor of Technology, Computer Science & Engineering GPA: 3.75 / 4.0</i>	<i>Aug 2019 – Jul 2023</i>

CERTIFICATIONS

React Native – The Practical Guide [2023]	Udemy
Database Design and Basic SQL in PostgreSQL	Coursera
Intermediate PostgreSQL	Coursera
Supervised Machine Learning: Regression and Classification	Coursera / DeepLearning.AI